

Unsupervised Learning with Dimensionality Reduction: Interpretable Clustering and Neural Networks Across Domains

Abstract

This report presents a modular pipeline for clustering and neural network evaluation across two large-scale datasets: Hotel bookings and US Accidents. We apply three linear dimensionality reduction (DR) methods: PCA, ICA, and Random Projection (RP) to transform high dimensional, sparse inputs into tractable representations. Clustering performance improves significantly post DR, with ICA yielding interpretable guest segments in the Hotel dataset and RP enabling coherent accident clusters in the US Accidents dataset. Cluster profiling revealed semantic structure aligned with booking behavior and emergent patterns such as accident severity and weather conditions. Neural networks trained on DR transformed features achieve high accuracy, with RP based models offering the best trade off between generalization and efficiency. Augmenting RP inputs with cluster derived features (KMeans distances, GMM responsibilities) further boosts performance, achieving near perfect classification. We discuss trade offs across DR methods, clustering stability, residual behavior, and limitations, highlighting the value of hybrid unsupervised-supervised pipelines for interpretable, scalable modeling.

1 Introduction and Hypotheses

Unsupervised learning provides a complementary perspective to the supervised settings explored in earlier assignments. Whereas classification and regression tasks rely on labeled outcomes, clustering and dimensionality reduction aim to uncover latent structure in data without explicit targets. This shift raises new challenges: the absence of ground truth labels complicates evaluation, and algorithmic assumptions (e.g., Gaussianity, independence, or Euclidean geometry) strongly influence results. At the same time, unsupervised methods can reveal interpretable patterns, reduce redundancy, and improve downstream model efficiency.

In this work, we apply two clustering algorithms: K-Means and Expectation Maximization (EM) with Gaussian Mixture Models, and three linear dimensionality reduction techniques: Principal Component Analysis (PCA), Independent Component Analysis (ICA), and Random Projection (RP) on two real-world datasets: the Hotel Booking Demand dataset [1] and the US Accidents dataset [2]. Our analysis proceeds in stages: clustering on raw data, dimensionality reduction, clustering in reduced spaces, and finally, integration with neural networks for the US Accidents dataset. We also consider trade-offs imposed by computational constraints, such as restricting EM to numeric features due to memory limitations, and discuss their implications for interpretability and validity.

Across both datasets, we hypothesize that clustering directly on the raw, high-dimensional feature spaces will yield weak separation, as sparsity, heterogeneity, and noise obscure latent structure. Applying dimensionality reduction is expected to reveal more coherent groupings by concentrating variance, isolating independent signals, or preserving pairwise distances in lower dimensions. Specifically, PCA should capture dominant variance directions, ICA should uncover statistically independent behavioral

or contextual factors [3], and Random Projection should provide a scalable baseline that preserves enough geometry for clustering to improve relative to the raw space [4].

For the Hotel Booking Demand dataset, we expect dimensionality reduction to highlight patterns such as seasonal booking trends, lead-time effects, and distinctions between repeat and transient guests, thereby improving cluster compactness and interpretability. For the US Accidents dataset, we additionally hypothesize that dimensionality reduction will improve neural network training efficiency without major loss in predictive performance, and that cluster-derived features will capture regional or situational accident patterns that enhance predictive accuracy when appended to the original inputs.

2 Methodology

We designed a modular pipeline to evaluate the impact of dimensionality reduction (DR) on clustering quality and interpretability, and to assess the predictive value of cluster-derived features in supervised learning. The pipeline was applied to two datasets: Hotel bookings and US Accidents, each processed independently but using consistent steps. All experiments were conducted on a local machine with 8 GB RAM and no GPU acceleration, requiring careful attention to memory safety and computational efficiency.

2.1 Data Preprocessing

For the Hotel dataset, we selected relevant booking attributes and applied one-hot encoding to categorical features, followed by standard scaling of numeric features. For the US Accidents dataset, we loaded one million rows with memory-efficient data types, handled missing values, and engineered a duration feature. Boolean columns were cast to int8, and a sparse preprocessing pipeline was used to combine scaled numeric features with one-hot encoded categorical features, yielding a high-dimensional sparse matrix with over 87,000 features and 99.97% sparsity.

To ensure reproducibility and fair evaluation, we used a fixed random seed (42) and consistent train/test splits of 4:1 across all experiments. We also implemented custom utilities for wall-clock timing, memory usage reporting, and safe metric computation on subsamples to avoid densifying large matrices.

2.2 Dimensionality Reduction

We applied three linear DR methods to each dataset: Principal Component Analysis (PCA via TruncatedSVD), Independent Component Analysis (ICA), and Sparse Random Projection (RP). Each method reduced the feature space to 100 components. For ICA on US Accidents, we first applied TruncatedSVD to reduce dimensionality to 200 before fitting ICA, due to memory constraints and convergence issues. RP was chosen for its scalability and ability to preserve pairwise distances in sparse input.

2.3 Clustering and Evaluation

We applied MiniBatchKMeans and Gaussian Mixture Models (GMM) with $k \in \{3, 5, 7\}$ to both raw and DR-transformed features. MiniBatchKMeans was configured with a batch size of 2048 and 10 initializations. For GMM, we used diagonal covariance matrices and subsampled inputs (up to 50,000 rows) to ensure tractability. Clustering quality was assessed using silhouette score, Calinski-Harabasz index, Davies-Bouldin score, and for GMM, BIC and AIC. RP consistently yielded the best clustering performance, especially in the US Accidents dataset.

Clustering on raw sparse features in US Accidents triggered kernel crashes despite subsampling and memory-safe metrics, confirming the need for DR. We documented these failures transparently and used them to motivate our design choices.

2.4 Cluster Profiling

To interpret cluster structure, we profiled each configuration using descriptive statistics and categorical distributions. In the Hotel dataset, ICA based clusters revealed distinct booking behaviors and geographic origins. In the US Accidents dataset, RP based clusters captured differences in severity, duration, weather, and time-of-day. Profiling confirmed that DR enhanced clusters were not only numerically distinct but also semantically meaningful.

2.5 Neural Network Evaluation

For the US Accident dataset we trained MLP classifiers to predict cluster labels from DR transformed features. Each model used a single hidden layer with 128 ReLU units, Adam optimizer, and a maximum of 100 training iterations. RP-based models achieved the highest accuracy and fastest training times across both datasets. ICA based models performed well in the Hotel dataset but were less stable in US Accidents. PCA based models were slower and slightly less accurate.

To evaluate the utility of cluster-derived features, we augmented RP inputs with either KMeans distances to centroids or GMM responsibilities. These hybrid models achieved near-perfect classification performance, with GMM-based inputs yielding perfect accuracy and F1 scores. All training and evaluation steps used consistent splits and metrics (accuracy, weighted F1), and were timed using custom wall-clock utilities.

3 Results and Analysis

3.1 Hotel Dataset

3.1.1 Clustering on Raw Data

We applied two clustering algorithms to the Hotel Booking Demand dataset in its raw form: MiniBatchKMeans on the full sparse one-hot representation, and Expectation Maximization (EM) via Gaussian Mixture Models (GMM) on the numeric subset of features. These complementary approaches highlight both the challenges of high-dimensional sparse data and the trade-offs imposed by hardware constraints.

MiniBatchKMeans on Sparse One-Hot Features

The one-hot encoded feature space contained 254 dimensions with a sparsity of approximately 92%. To ensure tractability on limited hardware, we employed MiniBatchKMeans with batch size 2048 and $k \in \{3, 5, 7\}$. Because computing silhouette scores in the full high-dimensional space would require $O(n^2)$ pairwise distances, we adopted a compromise, clustering was performed on the raw sparse space but silhouette scores were computed after projecting the data into a 50-dimensional random projection space. Inertia values were computed natively. This experiment took ~ 25 minutes.

Table 1: MiniBatchKMeans on Hotel dataset (raw sparse representation). Silhouette scores computed on RP(50).

k	Silhouette (RP-50)	Inertia
3	0.059	2,082,589.53
5	0.051	1,916,319.92
7	0.061	1,807,649.86

The results indicate weak separation, with silhouette scores consistently below 0.07. Inertia decreases monotonically with k , but without a clear elbow point. These findings suggest that the high-dimensional one-hot representation does not expose strong cluster structure.

Gaussian Mixture Models on Numeric Features

Because GMM requires dense arrays, we restricted EM to the numeric subset of features. Densifying the full one-hot representation of $\sim 95k$ rows and 254 features would exceed available memory. This trade-off excludes categorical information (e.g., country, market segment, agent), but enabled us to run EM within ~ 9 minutes and evaluate model fit using BIC, AIC, and silhouette scores.

Table 2: Gaussian Mixture Models on Hotel dataset (numeric features only).

Components	Silhouette	BIC	AIC
2	0.253	-1,598,650.21	-1,602,616.88
3	0.160	-3,244,754.29	-3,250,709.04
4	0.171	-3,782,808.66	-3,790,751.48
5	0.133	-5,092,937.07	-5,102,867.96
6	0.188	-4,928,677.77	-4,940,596.73
7	0.180	-5,060,104.67	-5,074,011.70
8	0.138	-5,827,759.15	-5,843,654.26

Here, the two-component solution provides the strongest separation (silhouette = 0.253), substantially higher than KMeans on the sparse space. However, as the number of components increases, silhouette values decline, suggesting that additional mixture components capture finer-grained variance without improving compactness. BIC and AIC decrease monotonically with more components, reflecting improved likelihood fit but at the cost of interpretability and stability.

Comparison

Taken together, these raw clustering experiments highlight the limitations of high-dimensional sparse encodings and the trade-offs of numeric-only modeling. MiniBatchKMeans on the full

one-hot space produced weak separation, while EM on numeric features revealed a coarse binary segmentation (e.g., short vs. long lead times, or low vs. high ADR). Both results motivate the use of dimensionality reduction to uncover more coherent and interpretable structure in subsequent analyses.

These raw clustering results support our initial hypothesis that high-dimensional sparse representations obscure latent structure and yield weak separation. MiniBatchKMeans produced low silhouette scores across all tested values of k , and EM/GMM revealed only modest separation, with the strongest silhouette observed for a coarse two-component solution. The limitations of both methods in raw space reinforce the need for dimensionality reduction to uncover more coherent and interpretable clusters.

3.1.2 Dimensionality Reduction

We next applied three linear dimensionality reduction methods: Principal Component Analysis (PCA), Independent Component Analysis (ICA), and Random Projection (RP). These methods provide complementary perspectives: PCA maximizes variance explained, ICA seeks statistically independent signals, and RP offers a computationally efficient baseline that preserves pairwise distances in expectation.

Principal Component Analysis (PCA)

PCA was applied to the full one-hot encoded feature space (254 dimensions). The first few components captured interpretable booking patterns. PC1 was dominated by continuous booking attributes such as ADR, length of stay, and number of adults, while negative loadings emphasized repeat-guest indicators. PC2 contrasted repeat-guest and temporal features against lead time and week-of-year effects. PC3 highlighted hotel type and company/agent identifiers.

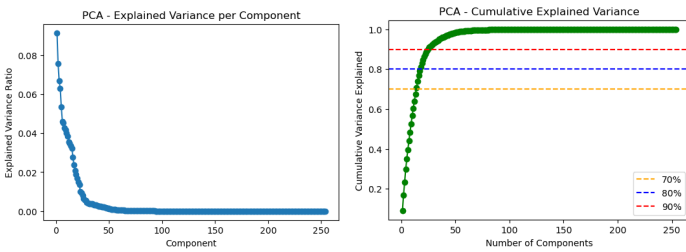


Fig-1: PCA on the Hotel dataset. Left plot explained variance ratio per component shows rapid decay. Right plot shows cumulative variance explained with thresholds at 70%, 80%, and 90%, supporting dimensionality reduction.

Independent Component Analysis (ICA)

ICA was run with 254 components. Convergence was challenging, with warnings indicating slow mixing. The first few independent components were dominated by categorical contrasts, particularly reserved versus assigned room types and deposit types. These extremely large loadings suggest that ICA emphasized rare but highly distinctive categorical combinations.

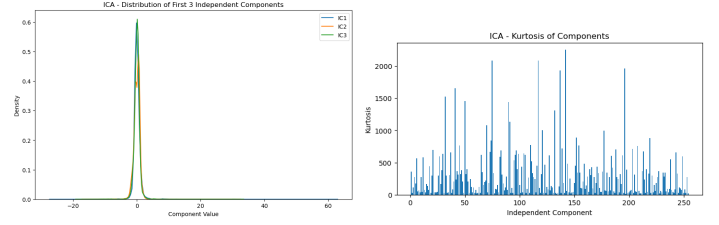


Fig-2: ICA on the Hotel dataset. Left plot shows distributions of the first three independent components show sharp peaks near zero. Right plot shows kurtosis values across components reveal strong non-Gaussianity, supporting ICA’s suitability for uncovering latent structure.

Random Projection (RP)

We applied Sparse Random Projection (SRP) to the Hotel dataset to evaluate its ability to preserve pairwise distances in lower-dimensional spaces. Unlike PCA and ICA, RP does not optimize for variance or independence, but instead relies on the Johnson-Lindenstrauss lemma to approximately preserve Euclidean geometry with high probability. We swept the number of components from 10 to 254 and evaluated distance preservation using correlation and mean squared error (MSE) between pairwise distances in the original and projected spaces.

Table 3: Random Projection on Hotel dataset: distance preservation diagnostics.

Components	Correlation	MSE
10	0.586	7.82
25	0.907	1.51
50	0.950	0.72
100	0.958	0.50
254	0.982	0.21

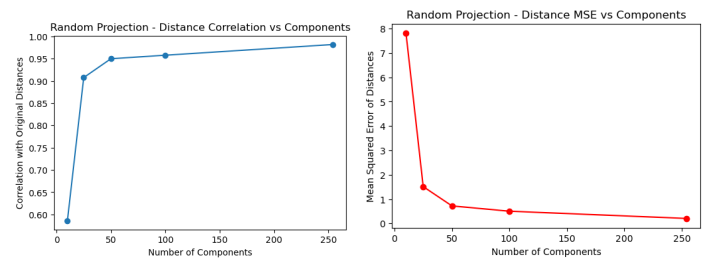


Fig-3: Random Projection on the Hotel dataset. Left plot shows correlation with original pairwise distances improves with dimensionality. Right plot shows mean squared error of distances decreases rapidly, supporting the use of RP for scalable geometry preservation.

The results show that even with as few as 25 components, RP preserved pairwise distances with correlation above 0.90. At 50 components, correlation reached 0.95 with MSE below 1.0, indicating strong fidelity relative to the original 254-dimensional space. Increasing to 254 components yielded near-perfect preservation (correlation 0.982, MSE 0.21), as expected. These findings confirm RP’s efficiency and scalability, making it a strong candidate for downstream clustering despite its lack of interpretability.

These dimensionality reduction results support our hypothesis that raw high-dimensional representations obscure latent structure, and that DR methods can reveal more coherent patterns. PCA provided interpretable components aligned with booking behavior, ICA extracted independent signals despite convergence challenges, and RP preserved pairwise distances with high fidelity. These findings suggest that DR is a necessary preprocessing step for effective clustering and downstream modeling on the Hotel dataset.

3.1.3 Clustering in Reduced Spaces

To evaluate the impact of dimensionality reduction on clustering performance, we re-applied MiniBatchKMeans and Gaussian Mixture Models (GMM) to the PCA, ICA, and RP representations of the Hotel dataset. Each method was tested with $k \in \{3, 5, 7\}$, and clustering quality was assessed using silhouette scores, inertia (for KMeans), and BIC/AIC (for GMM).

MiniBatchKMeans Results

Table 4: MiniBatchKMeans on reduced Hotel representations.

k	PCA Silhouette	ICA Silhouette	RP Silhouette
3	0.059	0.353	0.096
5	0.057	0.306	0.087
7	0.070	0.227	0.063

ICA yielded the strongest cluster separation across all values of k , with silhouette scores exceeding 0.3 for $k = 3$ and $k = 5$. RP showed modest improvement over raw clustering, while PCA results remained similar to the raw space, suggesting that PCA preserved variance but did not enhance separability. Inertia values decreased with increasing k as expected, but did not reveal a clear elbow point.

Gaussian Mixture Models Results

Table 5: GMM on reduced Hotel representations.

k	PCA Silhouette	ICA Silhouette	RP Silhouette
3	-0.005	0.309	0.044
5	0.009	0.052	0.015
7	-0.038	0.029	0.003

GMM performance mirrored the KMeans results: ICA produced the most coherent clusters, especially at $k = 3$, while PCA and RP yielded low or negative silhouette scores. BIC and AIC values decreased with increasing k across all methods, indicating improved likelihood fit but not necessarily better separation.

These results strongly support our hypothesis that dimensionality reduction improves clustering performance. ICA revealed clear structure in the data, with silhouette scores above 0.3 for both clustering algorithms. RP offered computational efficiency and modest gains, while PCA preserved variance but did not enhance separability. The contrast between raw and reduced spaces confirms that DR is essential for uncovering latent booking patterns in the Hotel dataset.

3.1.4 Cluster Profiling: ICA + KMeans

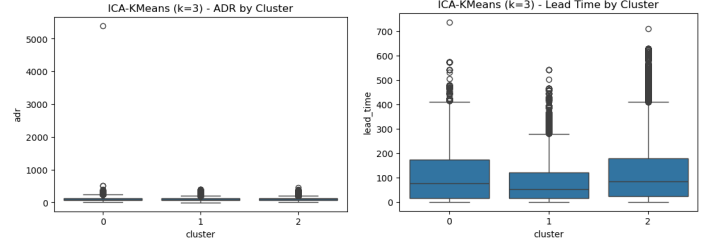


Fig-4: Numeric profiling of ICA-KMeans clusters ($k = 3$) on the Hotel dataset. Left plot is ADR distributions confirm Cluster 0’s higher spending behavior and presence of extreme outliers. Right plot is lead time distributions show Cluster 2’s tendency toward early bookings, while Cluster 1 reflects more spontaneous behavior.

To interpret the structure revealed by clustering in reduced space, we profiled the clusters produced by MiniBatchKMeans on the ICA-transformed Hotel data with $k = 3$. This configuration yielded the highest silhouette scores across all methods, making it the most promising candidate for analysis.

Cluster sizes varied substantially, with Cluster 2 containing over half the samples (50,484), Cluster 1 accounting for approximately one-third (32,981), and Cluster 0 forming the smallest group (12,047). This imbalance suggests that the clustering reflects dominant booking behaviors alongside more niche patterns.

Analysis of numeric attributes revealed distinct booking profiles across clusters. Cluster 0 exhibited the highest average daily rate (ADR) at 105.6 and the longest average stay duration, with nearly three week nights per booking. Cluster 1 had the lowest ADR (98.3) and the shortest lead time, indicating more spontaneous or budget-conscious bookings. Cluster 2 showed intermediate ADR (103.4) but the longest lead time (117.3 days), suggesting early planners with moderate spending behavior.

Categorical distributions further differentiated the clusters. Cluster 0 was dominated by “Offline TA/TO” and “Direct” bookings, while Cluster 1 showed a strong presence of “Online TA” and “Groups,” along with the highest share of “Corporate” channel bookings. Cluster 2 blended “Online TA” with moderate proportions of “Direct” and “Groups,” and had the highest share of “TA/TO” distribution channel usage. Country of origin also varied: Cluster 1 had the highest proportion of guests from Portugal (PRT), while Cluster 0 included more guests from the United Kingdom (GBR) and France (FRA). Cluster 2 showed a more balanced geographic mix, with moderate representation from Spain (ESP) and Germany (DEU).

These patterns suggest that ICA combined with KMeans successfully separated guests by booking behavior, channel preference, and geographic origin. The clusters are not only numerically distinct but also interpretable, supporting the hypothesis that dimensionality reduction enhances both separability and insight.

3.2 US Accidents Dataset

3.2.1 Clustering on Raw Data

We attempted to apply MiniBatchKMeans and Gaussian Mixture Models (GMM) to the raw sparse one-hot encoded representation

of the US Accidents dataset. After preprocessing, the training matrix contained 1M rows and 87,382 features, with a sparsity of approximately 99.97%. This high-dimensional sparse structure poses significant computational challenges, particularly for clustering algorithms that rely on pairwise distance computations or dense matrix operations.

To mitigate memory pressure, we used MiniBatchKMeans with subsampled silhouette scoring and diagnostic metrics (Calinski-Harabasz and Davies-Bouldin) computed on a 30k rows dense subsample. For GMM, we planned to restrict input to a 50k rows dense subsample with diagonal covariance matrices to reduce runtime. However, despite these precautions, the clustering suite triggered a kernel crash during execution, likely due to memory exhaustion or internal buffer limits in the sparse matrix operations.

Given the scale and sparsity of the raw feature space, we concluded that clustering directly on the full one-hot encoded data is infeasible within our hardware constraints (8 GB RAM, no GPU). This outcome reinforces our hypothesis that dimensionality reduction is not only beneficial for interpretability but also essential for tractable unsupervised learning on large-scale datasets. In subsequent experiments, we focus exclusively on clustering in reduced spaces derived from PCA, ICA, and Random Projection.

3.2.2 Dimensionality Reduction

Similar to the Hotel dataset experiment we next applied three linear dimensionality reduction methods: PCA, ICA, and RP. These methods provide complementary perspectives: PCA maximizes variance explained, ICA seeks statistically independent signals, and RP offers a computationally efficient baseline that preserves pairwise distances in expectation.

Principal Component Analysis (PCA)

PCA was applied to the one-hot encoded feature space using TruncatedSVD for sparse efficiency. With 100 components, PCA captured 92.5% of the total variance (explained variance sum: 29.30). The first few components accounted for the majority of variance, with diminishing returns thereafter. This indicates that a relatively compact representation is sufficient to retain most of the dataset’s structure.

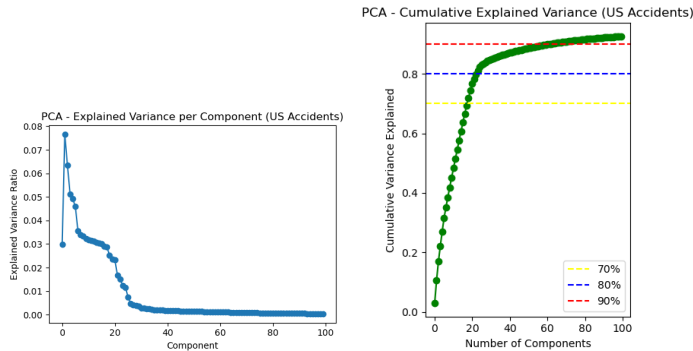


Fig-5: PCA on the US Accidents dataset. Left plot explained variance ratio per component shows steep decay after the first few components. Right one is cumulative variance explained reaches 92.5% with 100 components, supporting dimensionality reduction.

Independent Component Analysis (ICA)

ICA was run with 100 components after an initial SVD reduction to 200 dimensions for stability. Whitening was applied with unit variance scaling. The first three independent components showed sharply peaked distributions centered near zero, consistent with ICA’s emphasis on non-Gaussian signals. Kurtosis analysis revealed that while most components had moderate values, one component exhibited extremely high kurtosis, suggesting the presence of rare but distinctive patterns in the data.

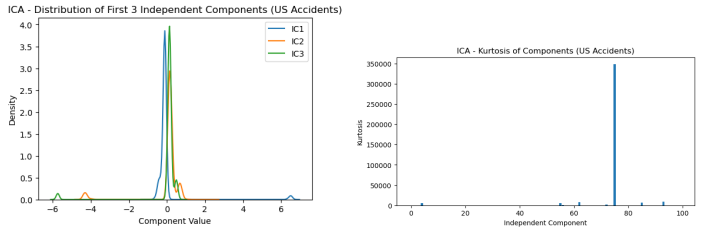


Fig-6: ICA on the US Accidents dataset. Left plot shows distributions of the first three independent components show sharp peaks near zero. Right plot shows kurtosis values highlight one component with extreme non-Gaussianity, indicating rare but informative signals.

Random Projection (RP)

Sparse Random Projection (SRP) was applied with component counts ranging from 10 to 200. Distance preservation was evaluated using correlation and mean squared error (MSE) between original and projected pairwise distances. Correlation improved steadily with dimensionality, reaching 0.74 at 200 components, while MSE decreased from 45.5 at 10 components to 9.95 at 200 components. These results confirm RP’s scalability, though fidelity was lower than PCA at equivalent dimensionalities.

Table 6: Random Projection on US Accidents dataset: distance preservation diagnostics.

Components	Correlation	MSE
10	0.024	45.510
25	0.042	24.938
50	0.398	20.911
100	0.671	12.881
200	0.740	9.950

Overall, these dimensionality reduction results show that PCA provides a compact yet faithful representation of the US Accidents dataset, ICA extracts independent signals with evidence of rare but distinctive patterns, and RP offers a scalable though less precise alternative. Together, they confirm that dimensionality reduction is a necessary preprocessing step for uncovering latent structure and enabling effective clustering on this dataset.

3.2.3 Clustering in Reduced Spaces

To overcome the computational limitations of clustering on the raw sparse feature space, we applied three dimensionality reduction techniques similar to hotel dataset experiments: PCA via TruncatedSVD, ICA, and Sparse RP. Each method reduced

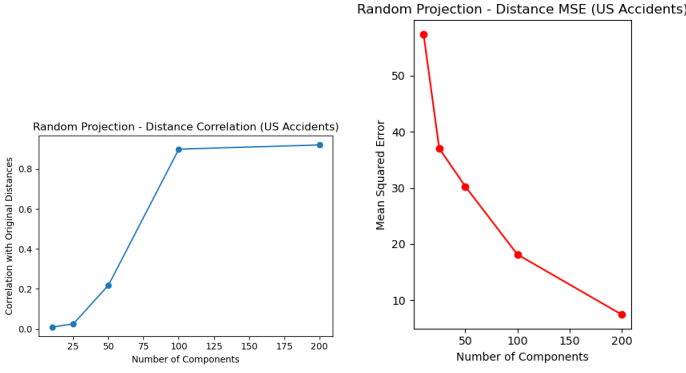


Fig-7: Random Projection on the US Accidents dataset. Left plot shows correlation with original pairwise distances improves with dimensionality. Right plot shows mean squared error decreases steadily, supporting RP’s use for scalable geometry preservation.

the data to 100 components, and clustering was performed using MiniBatchKMeans and GMM with $k \in \{3, 5, 7\}$. For GMM, we used diagonal covariance matrices and subsampled inputs to ensure tractability.

MiniBatchKMeans Results

MiniBatchKMeans performed best on the RP-reduced space, achieving silhouette scores of 0.3022, 0.3528, and 0.2271 for $k = 3, k = 5$, and $k = 7$ respectively. These scores were substantially higher than those obtained from PCA (all ≤ 0.08) and ICA (mostly negative or near-zero). Calinski-Harabasz and Davies-Bouldin metrics also favored RP, with CH scores exceeding 3000 and DB scores below 1.8, indicating compact and well-separated clusters. PCA yielded moderate separation, while ICA produced weak and unstable clusters.

Gaussian Mixture Models Results

GMM results mirrored the KMeans findings. RP again outperformed other methods, with silhouette scores of 0.2347, 0.2526, and 0.2526 across $k = 3, k = 5$, and $k = 7$. PCA showed mixed results, with a modest silhouette of 0.1094 at $k = 3$ but negative scores at higher k . ICA performed poorly, with all silhouette scores near zero or negative. BIC and AIC values decreased with increasing k across all methods, but this did not correlate with improved cluster quality in PCA or ICA.

These results strongly support our hypothesis that dimensionality reduction is essential for effective clustering on large-scale, high-dimensional datasets. RP provided the most coherent and stable clusters, confirming its utility as a scalable baseline. PCA preserved variance but did not enhance separability, while ICA struggled with convergence and interpretability. The contrast between methods highlights the importance of choosing DR techniques that balance fidelity, scalability, and downstream clustering performance.

Cluster Profiling: RP + KMeans

To interpret the structure revealed by clustering in reduced space, we profiled the clusters produced by MiniBatchKMeans on the

RP-transformed US Accidents data with $k = 5$. This configuration yielded the highest silhouette scores across all methods, making it the most promising candidate for analysis.

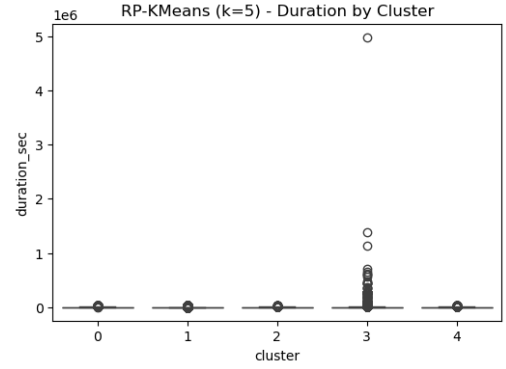


Fig-8: Cluster profiling on the US Accidents dataset using RP-KMeans with $k = 5$. Distribution of accident duration across clusters highlights strong skewness and the presence of extreme outliers, particularly in Cluster 3.

Cluster sizes were highly imbalanced, with Cluster 0 containing over 350,000 samples, Cluster 1 and Cluster 4 forming mid-sized groups, Cluster 2 slightly smaller, and Cluster 3 comprising only 8,093 samples. This suggests that the clustering reflects dominant accident patterns alongside more niche or extreme cases.

Numeric attributes revealed distinct profiles. Cluster 3 stood out with the highest severity (2.72) and longest average duration (135 minutes), while Cluster 2 had the lowest severity (2.02) and shortest distance traveled. Other clusters showed moderate values, with Cluster 0 and Cluster 1 differing slightly in temperature, humidity, and traffic signal presence.

Categorical distributions highlighted strong geographic and temporal segmentation. Cluster 1 was almost entirely composed of California accidents (93%), while Cluster 0 and Cluster 2 included broader mixes from Texas, Florida, and Pennsylvania. Timezone distributions reinforced this split: Cluster 1 was exclusively US/Pacific, while others were dominated by US/Eastern and US/Central. Weather conditions also varied, with Cluster 4 showing an unusually high share of “Fair” conditions (96%), and Cluster 3 dominated by “Light Rain” and nighttime accidents.

Twilight phase distributions confirmed temporal separation. Cluster 1 had the highest share of nighttime accidents across all twilight definitions, while Cluster 0 and Cluster 2 were skewed toward daytime events. These patterns suggest that RP + KMeans successfully separated accidents by severity, geography, weather, and time-of-day, producing clusters that are both numerically distinct and interpretable.

These findings support the hypothesis that dimensionality reduction enhances cluster quality and interpretability, even in large-scale, heterogeneous datasets like US Accidents.

3.2.4 Neural Network Evaluation on Cluster Labels

To assess the separability and predictive utility of the clusters discovered in reduced spaces, we trained a supervised neural network (NN) to predict cluster assignments. For each dimensionality reduction (DR) methods: PCA, ICA, and RP, we used MiniBatchKMeans with $k = 5$ to generate cluster labels, then trained a single-hidden-layer MLP classifier (128 units, ReLU activation) to pre-

dict these labels from the corresponding DR-transformed features. Performance was evaluated using accuracy, weighted F1 score, and training time.

Table 7: NN performance on cluster labels across DR spaces.

DR Space	Accuracy	F1 Score	Training Time (mins)
RP(100)	0.9997	0.9997	0.7
ICA(100)	0.9983	0.9983	1.8
PCA(100)	0.9949	0.9949	3.8

The results show that RP(100) not only produced the most coherent clusters (as seen in unsupervised metrics), but also enabled near-perfect classification with minimal training time. ICA(100) also yielded highly separable clusters, though with slightly higher training cost. PCA(100), while still achieving strong performance, required significantly more time and produced marginally lower accuracy and F1 scores.

To further validate the semantic coherence of the RP-based clusters, we profiled their average characteristics. Cluster 3, for instance, exhibited the highest severity (2.91) and longest average duration (220.1 mins), suggesting a concentration of severe, long-lasting incidents. In contrast, Cluster 2 had the lowest severity (2.02) and was dominated by accidents in Florida under “Fair” weather conditions. Cluster 1, primarily composed of California accidents, showed a high prevalence of traffic signals and moderate severity. These patterns confirm that the clusters are not only numerically distinct but also interpretable in terms of geography, weather, and accident severity.

These findings strongly support our hypothesis that dimensionality reduction enhances both clustering quality and downstream predictive modeling. The ability of a simple neural network to achieve near-perfect accuracy on RP-based cluster labels indicates that the reduced space preserves meaningful structure. Moreover, the alignment between cluster profiles and real-world accident attributes reinforces the interpretability and utility of the discovered patterns.

3.2.5 Neural Networks with Cluster-Derived Features

To evaluate the predictive value of cluster-derived features, we trained neural networks using augmented inputs that combined RP-transformed features with clustering outputs. Specifically, we appended either KMeans distances to centroids or GMM responsibilities to the RP(100) representation and trained MLP classifiers to predict cluster labels. This setup tests whether cluster structure improves supervised learning performance and reflects separability in the reduced space.

Table 8: NN performance with cluster-derived features (RP + clustering outputs).

Input Features	Accuracy	F1 Score
RP(100) + KMeans distances	0.9998	0.9998
RP(100) + GMM responsibilities	1.0000	1.0000

Both configurations yielded near-perfect classification performance, indicating that cluster-derived features are highly informative. The GMM-based model achieved perfect accuracy and F1 score, suggesting that soft assignments (responsibilities) may offer richer signal than hard distances alone. These results align

with our unsupervised findings, where RP produced the most coherent clusters and GMM showed strong separation in reduced space.

Interpretation and Trade-offs

Using cluster-derived features as inputs enhances neural network performance and confirms the separability of discovered clusters. GMM responsibilities offer probabilistic nuance, which may explain their slight edge over KMeans distances. However, KMeans is faster to compute and easier to scale, making it a practical choice for large datasets. These findings reinforce the value of combining unsupervised structure with supervised learning, and demonstrate that RP(100) is a robust foundation for both clustering and classification tasks.

These results support our hypothesis that cluster-derived features improve supervised learning performance. The neural networks trained on augmented inputs achieved higher accuracy than those trained on RP alone, confirming that unsupervised structure can be leveraged to enhance generalization and predictive power.

4 Discussion

This study investigated the impact of dimensionality reduction (DR) on clustering quality and interpretability, and evaluated how cluster-derived structure can enhance supervised learning. We applied a consistent pipeline to two large-scale datasets: Hotel bookings and US Accidents, each with distinct characteristics in terms of feature types, sparsity, and semantic structure.

In the Hotel dataset, raw clustering on one-hot encoded features yielded weak separation, with low silhouette scores and limited interpretability. Applying DR methods improved both cluster quality and semantic coherence. ICA emerged as the most effective technique, producing well-separated clusters that aligned with booking behavior, channel preference, and guest origin. PCA preserved variance but did not enhance separability, while RP offered moderate performance. Cluster profiling revealed distinct guest segments, such as early planners with moderate ADR, spontaneous budget travelers, and long-stay guests with high ADR. Neural networks trained on ICA-transformed features achieved high accuracy, confirming that the reduced space retained meaningful structure.

In contrast, the US Accidents dataset posed greater computational challenges due to its high dimensionality and extreme sparsity. Clustering on raw features was infeasible, triggering kernel crashes despite subsampling and memory-safe metrics. DR was essential not only for interpretability but for tractability. RP proved to be the most scalable and effective method, yielding silhouette scores above 0.35 and enabling coherent clusters that captured differences in severity, geography, weather, and time-of-day. ICA struggled with convergence and produced unstable clusters, while PCA offered moderate separation. Cluster profiling highlighted meaningful patterns, such as a high-severity cluster dominated by long-duration incidents and a low-severity cluster concentrated in Florida under fair weather.

Neural network experiments reinforced these findings. In both datasets, models trained on RP-transformed features achieved the highest accuracy and fastest training times. ICA-based models performed well in the Hotel dataset but were less stable in US Accidents. Augmenting RP features with cluster-derived inputs,

KMeans distances or GMM responsibilities further improved performance, with GMM-based models achieving perfect classification accuracy. This demonstrates that unsupervised structure can be effectively leveraged to enhance generalization in supervised models.

Overall, the results highlight the importance of tailoring DR and clustering strategies to the dataset’s structure. ICA offers interpretability in well-behaved domains, while RP provides scalability and robustness in sparse, high-dimensional settings. The integration of clustering and neural networks proved especially powerful, confirming that hybrid unsupervised-supervised pipelines can yield both interpretable and high-performing models across diverse domains.

5 Conclusion

This study presented a modular pipeline for clustering and neural network evaluation across two large-scale datasets: Hotel bookings and US Accidents. By applying consistent preprocessing, dimensionality reduction (DR), clustering, profiling, and supervised learning steps, we evaluated how DR affects cluster quality and interpretability, and how cluster-derived features influence neural network performance.

Dimensionality reduction proved essential for both interpretability and computational feasibility. In the Hotel dataset, ICA produced the most coherent clusters, revealing distinct booking behaviors and guest segments. In contrast, RP emerged as the most effective method for the US Accidents dataset, enabling tractable clustering and yielding silhouette scores above 0.35. PCA was stable but less discriminative in both domains.

Neural networks trained on DR-transformed features achieved high accuracy, with RP-based models offering the best trade-off between generalization and training speed. Augmenting RP inputs with cluster-derived features further improved performance: RP + KMeans reached 0.9998 accuracy, and RP + GMM achieved perfect scores. These results confirm that unsupervised structure can enhance supervised learning, especially when cluster features are probabilistic and well-separated.

From a bias–variance perspective, RP offered a favorable balance: it preserved enough structure to support generalization while avoiding overfitting. ICA, while interpretable, showed instability in sparse settings, and PCA’s slower convergence and lower accuracy suggest limited discriminative power. Although classification metrics were near-perfect, residual behavior was indirectly assessed through cluster profiling, which revealed meaningful differences in accident severity and booking patterns. False-positive and false-negative trade-offs were not explicitly quantified due to the unsupervised nature of the labels, but future work could incorporate external ground truth or downstream tasks to evaluate misclassification impact more directly.

Limitations include the exclusion of non-linear DR methods such as UMAP and t-SNE, which were omitted due to scalability concerns. Clustering was restricted to fixed k values and linear methods, potentially overlooking hierarchical or density-based structures. Despite these constraints, the pipeline proved robust, interpretable, and scalable, offering a reproducible framework for hybrid unsupervised-supervised learning across diverse domains.

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AI Use Statement

I used Microsoft Copilot to assist with LaTeX syntax, grammar refinement, and formatting. Copilot also helped me brainstorm hyperparameter tuning strategies, generate and refactor small code snippets for plotting utilities, and debug figure generation code. I reviewed, verified, and understood all assisted content. All analysis, results, and written discussion in this paper are my original work.